Optimization Final Project: Digital Color Restoration

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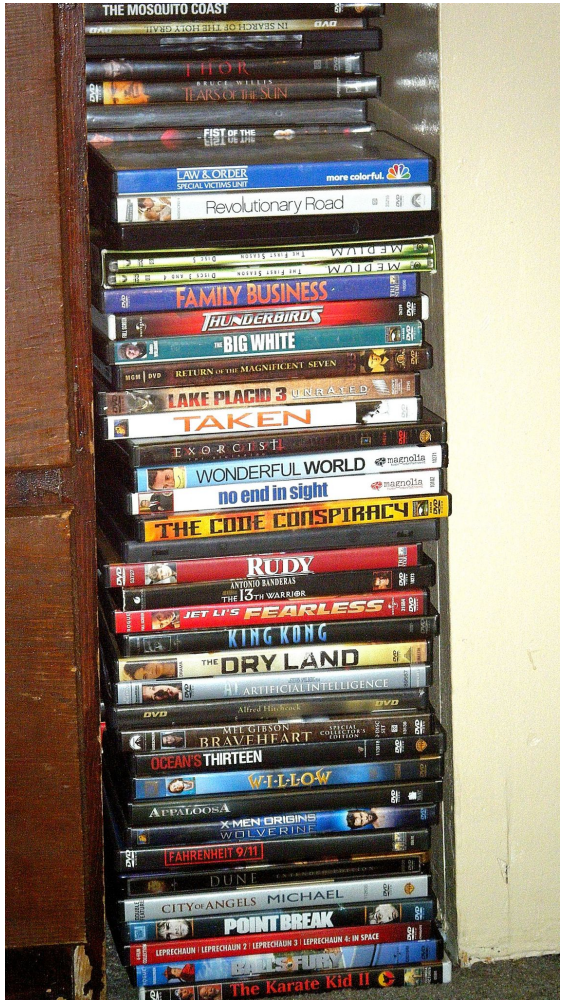
Introduction

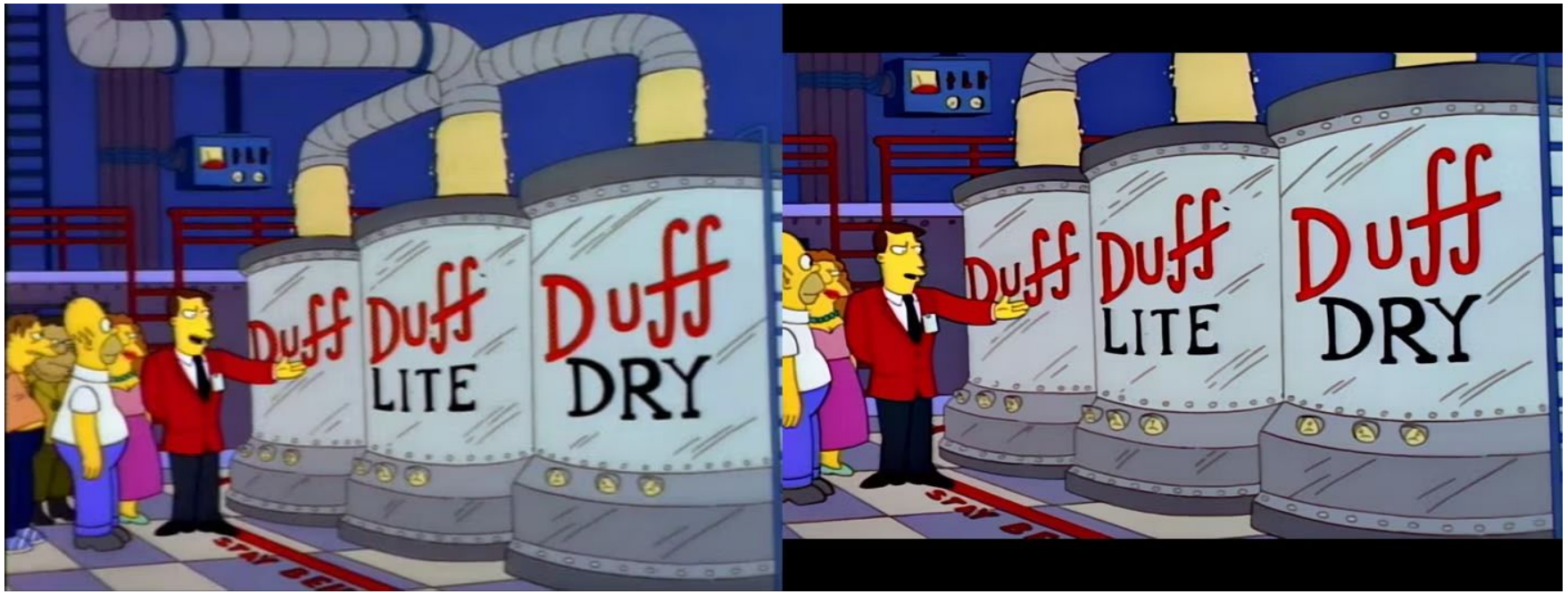
Physical Media

- Any source of imagery or video data that is tangible and analog (i.e. photographs, films, VHS, DVD, etc.)
- A prominent component to the state of multimedia especially during the age of streaming services
- Often provides a much better visual resolution than digital imagery due to being analog data, but often more subject to environmental circumstances



Remasters

- Typically done through taking the original film of a specific TV show or movie and running it through newer technology



Sailor Moon

- Created by Toei Animation in 1992 based on mangaka of the same name made by Naoko Takeuchi
- Had 200 episodes and 3 specials in its original run, but continues to get remakes in different media
- Massively successful both in Japan and International audiences, introducing many to Japanese animation
- Original physical media scans are at risk of decaying beyond recovery

HAS NEVER ACTUALLY WATCHED A SINGLE EPISODE OF SAILOR MOON

DAN
FAKE?

Film Degradation

- While film is built to last longer than other forms of physical media, it is also subject to decay if mishandled
- Film positives are much more sensitive to environmental conditions than film negatives
- The film used to store Sailor Moon was an Eastman Kodak 16mm film, which is notorious for its color degradation to red or green

For this project, I intend to use less-than-linear programming optimization to perform color restoration on digital images.

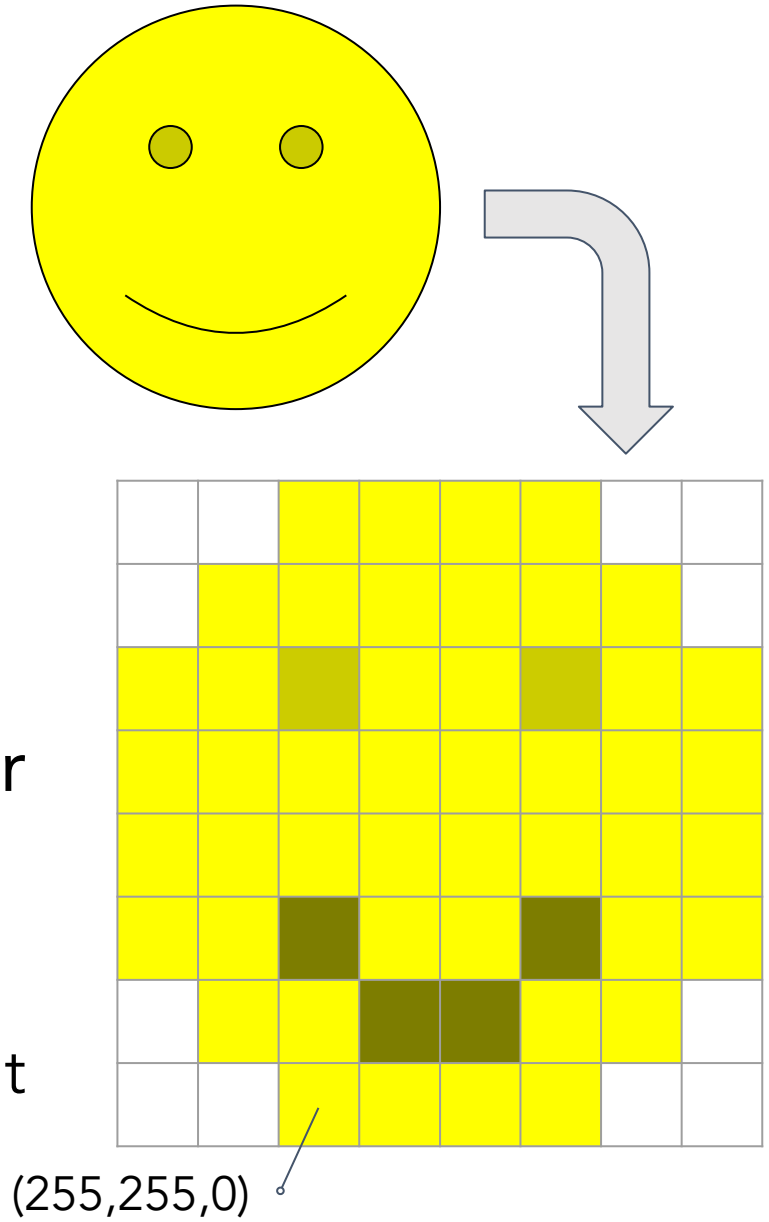
Additional Disclaimers:

- This optimization experiment is mostly for fun, I do not intend to be hired by Toei or Viz Media for image restoration on a professional basis
- The digital scans can be argued to also not be the "intended form" of viewing the media, as many cartoons at the time were designed to be viewed by a CRT screen, digital is just used to make computations easier
- Sailor Moon is property of Toei Animation and Viz Media, I do not any of the imagery depicted

Quick Overview of Computer Vision

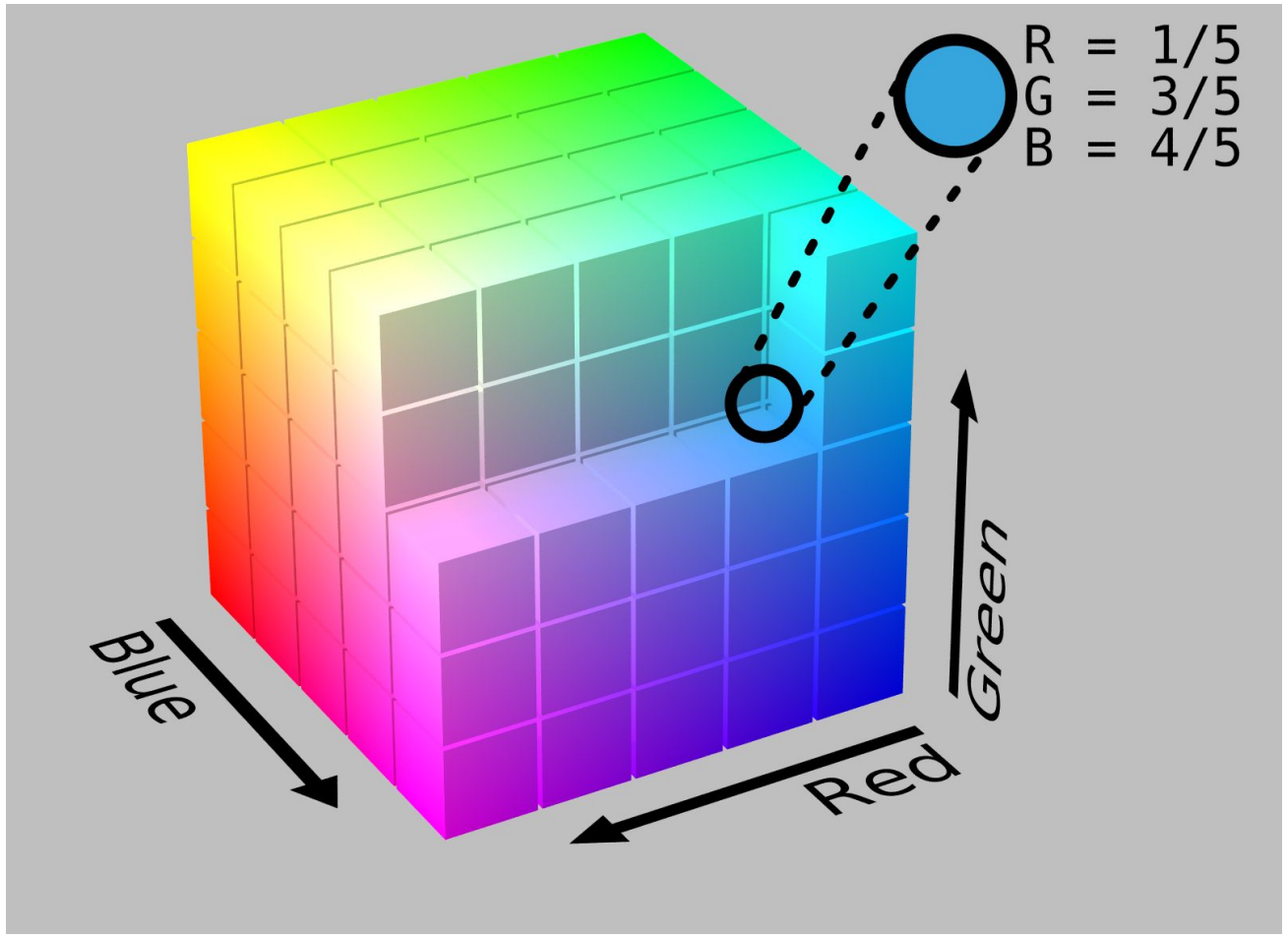
Digital Images

- Digital images are sampled from a lens or physical media into a series of discrete bytes representing illumination values known as "pixels"
- The number of rows and columns of pixels within a digital image represent its spatial resolution
- Within a color image, pixels represent color in three spectrum bands: Red, Green, and Blue
  - A pixel can vary in depth depending on the hardware, but the standard is typically valued at a byte (8-bit) or 256 possible values



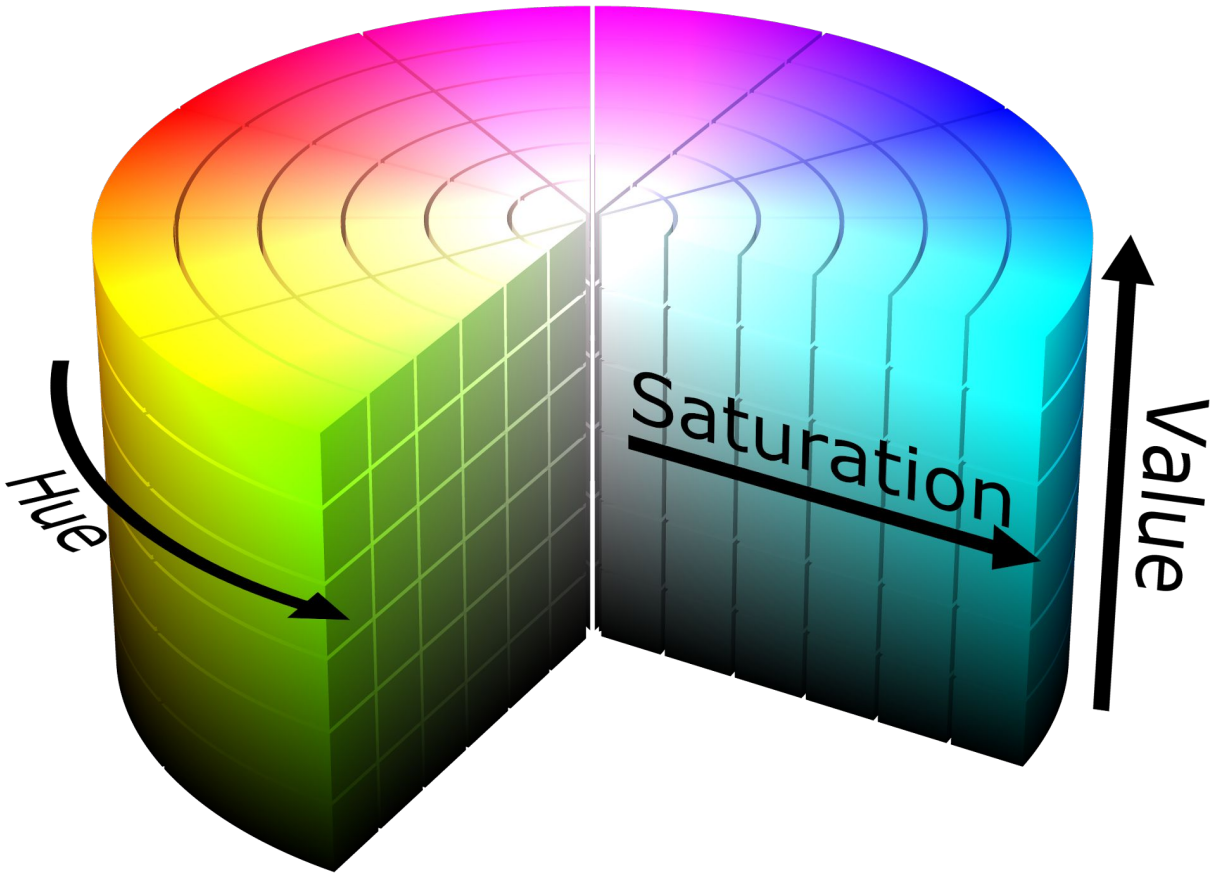
Color Spaces: RGB

- The standard color space
- Every color is a represented as a permutation of the colors Red, Green, and Blue
- In an byte-pixel depth representation, there are 16,777,216 possible colors able to be represented by the RGB color space
- The color space that all digital images are displayed using
- Has an inverse counterpart called CYMK or subtractive for the use of printing



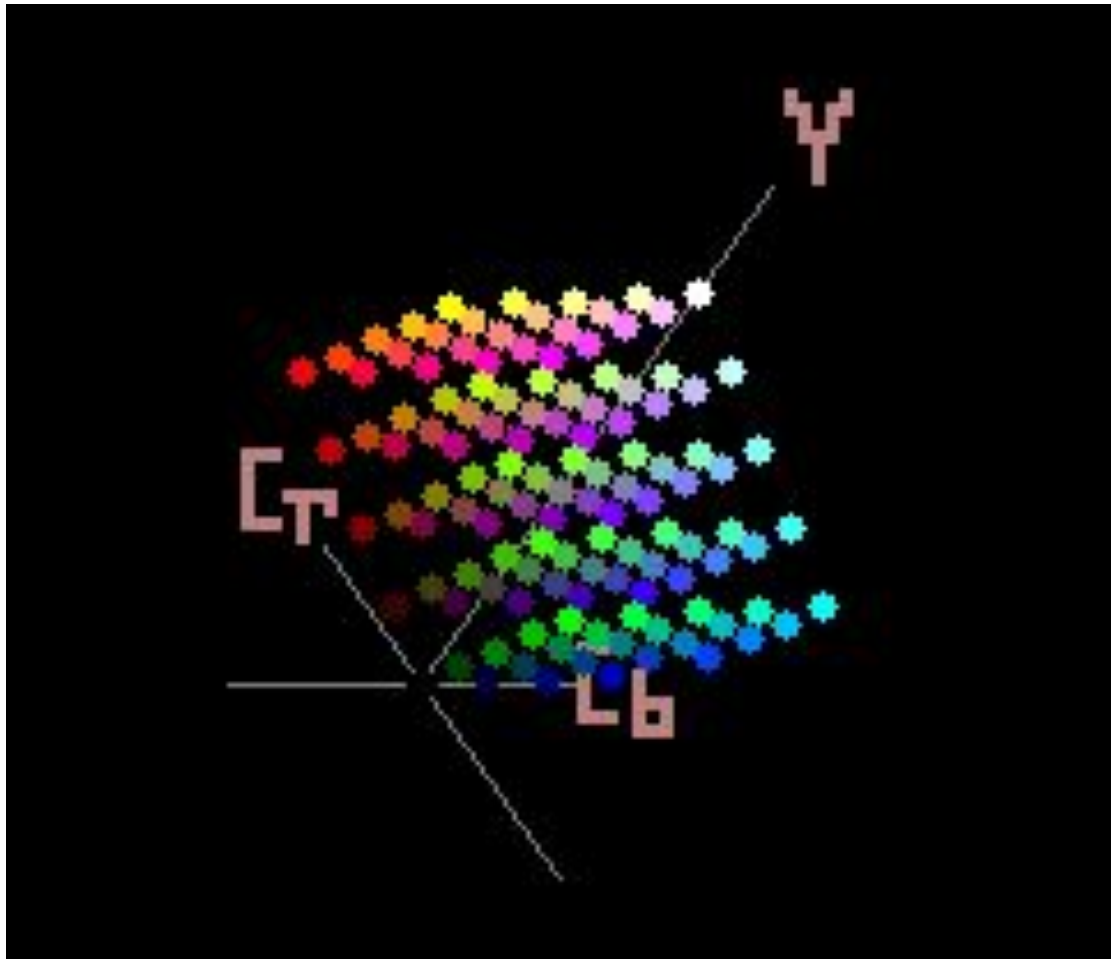
Color Spaces: HSV

- Developed to be more intuitive than the cartesian representation of RGB, and represents colors as a permutation of a Hue, Saturation, and Value/Intensity
- Hue is the angular dimensions containing the colors in a "color wheel" format
- Saturation is the chromatic intensity of a particular color from gray to a bright color
- Value is the luminous intensity of a particular color ranging from black to maximum brightness



Color Spaces: YCbCr

- Developed for storage and transmission of images to reduce the redundancy of RGB images
- Represents colors as a permutation of luminance or brightness and chrominance or color difference
- Cb represents blue-difference chrominance
- Cr represents red-difference chrominance



Optimization Design

Decision Variables

- Decision variables will appear as the scalar modifications to each of the color bands experimented on:
- $$I_{RGB}' = [d_R I_R, d_G I_G, d_B I_B]^T$$
  $$I_{HSV}' = [d_H I_H, d_S I_S, d_V I_V]^T$$
  $$I_{YCbCr}' = [d_Y I_Y, d_{Cb} I_{Cb}, d_{Cr} I_{Cr}]^T$$
- It is typically better to make decision variables that span for each pixel but that has a much longer computational load

Objective Function

$$\Delta E^*_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \frac{\Delta C'}{k_C S_C}\frac{\Delta H'}{k_H S_H}}$$

Where …

$$\Delta L' = L^*_2 - L^*_1$$

$$T = 1 - 0.17\cos(\bar{H}' - 30^\circ) + 0.24\cos(2\bar{H}') + 0.32\cos(3\bar{H}' + 6^\circ) - 0.20\cos(4\bar{H}' - 63^\circ)$$

$$\bar{L} = \frac{L^*_1 + L^*_2}{2} \quad \bar{C} = \frac{C^*_1 + C^*_2}{2} \quad \text{where } C^*_1 = \sqrt{a^{*2}_1 + b^{*2}_1}, \quad C^*_2 = \sqrt{a^{*2}_2 + b^{*2}_2},$$

$$\Delta h' = \begin{cases} h'_2 - h'_1 & |h'_1 - h'_2| \le 180^\circ \\ h'_2 - h'_1 + 360^\circ & |h'_1 - h'_2| > 180^\circ, h'_2 \le h'_1 \\ h'_2 - h'_1 - 360^\circ & |h'_1 - h'_2| > 180^\circ, h'_2 > h'_1 \end{cases}$$

$$a'_1 = a^*_1 + \frac{a^*_1}{2}\left(1 - \sqrt{\frac{\bar{C}^7}{\bar{C}^7 + 25^7}}\right) \quad a'_2 = a^*_2 + \frac{a^*_2}{2}\left(1 - \sqrt{\frac{\bar{C}^7}{\bar{C}^7 + 25^7}}\right)$$

$$\bar{C}' = \frac{C'_1 + C'_2}{2} \text{ and } \Delta C' = C'_2 - C'_1 \quad \text{where } C'_1 = \sqrt{a'^2_1 + b^{*2}_1} \quad C'_2 = \sqrt{a'^2_2 + b^{*2}_2}$$

$$S_L = 1 + \frac{0.015(\bar{L} - 50)^2}{\sqrt{20 + (\bar{L} - 50)^2}} \quad S_C = 1 + 0.045\bar{C}' \quad S_H = 1 + 0.015\bar{C}'T$$

$$h'_1 = \text{atan2}(b^*_1, a'_1) \mod 360^\circ, \quad h'_2 = \text{atan2}(b^*_2, a'_2) \mod 360^\circ$$

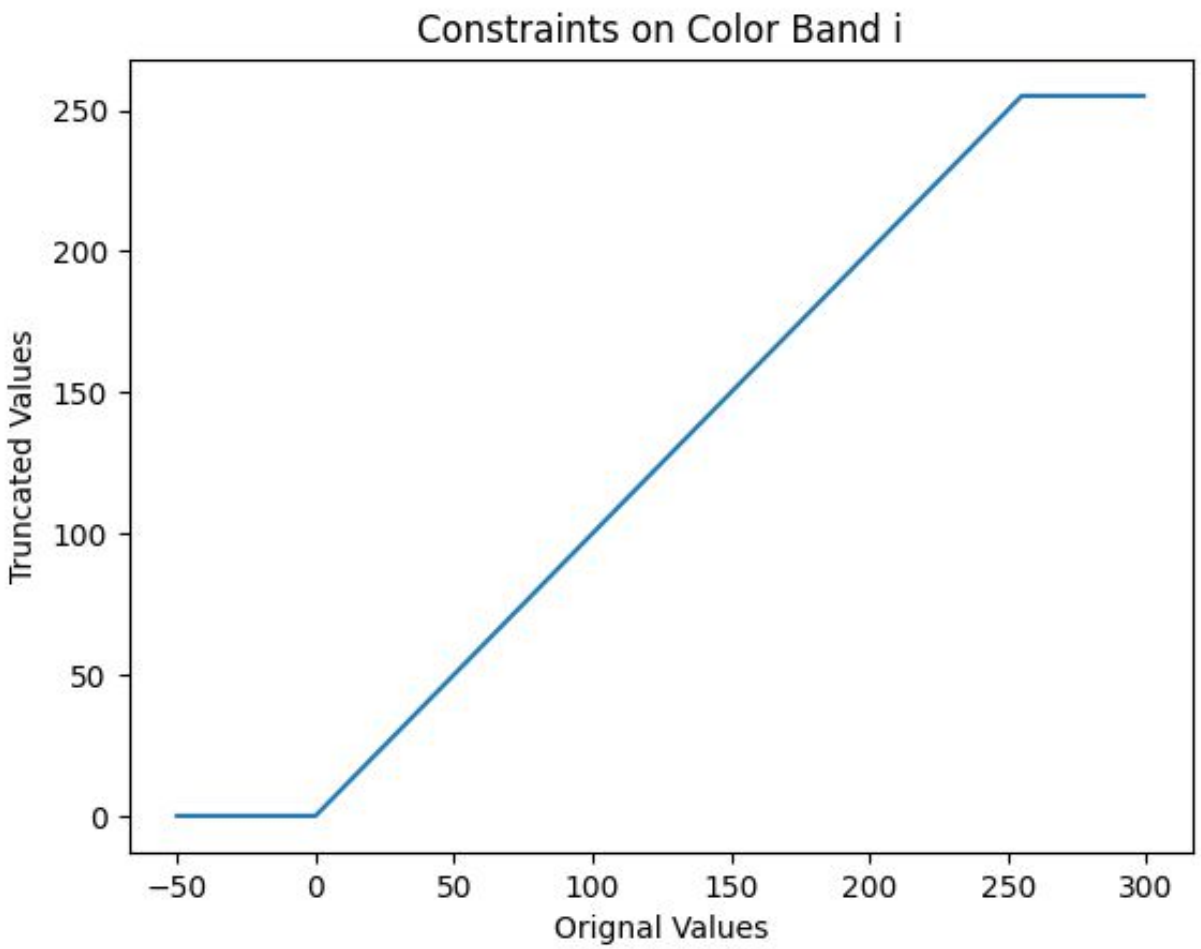
$$R_T = -2\sqrt{\frac{\bar{C}'^7}{\bar{C}'^7 + 25^7}} \sin\left[60^\circ \cdot \exp\left(-\left[\frac{\bar{H}' - 275^\circ}{25^\circ}\right]^2\right)\right]$$

$$\Delta H' = 2\sqrt{C'_1 C'_2}\sin(\Delta h'/2), \quad \bar{H}' = \begin{cases} (h'_1 + h'_2)/2 & |h'_1 - h'_2| \le 180^\circ \\ (h'_1 + h'_2 + 360^\circ)/2 & |h'_1 - h'_2| > 180^\circ, h'_1 + h'_2 < 360^\circ \\ (h'_1 + h'_2 - 360^\circ)/2 & |h'_1 - h'_2| > 180^\circ, h'_1 + h'_2 \ge 360^\circ \end{cases}$$

```
image1_lab = cv2.cvtColor(image_ref, cv2.COLOR_RGB2Lab)
image2_lab = cv2.cvtColor(image_recon, cv2.COLOR_RGB2Lab)
return np.sum(colour.delta_E(image1_lab, image2_lab, method='CIE 2000'))
```

Constraints

- Because we are dealing with image data, there is a value limitation between 0 and 255 (0 and 179 for H bands), but in the case of overflow, values can simply just wrap around
- Instead of a traditional constraint, data is truncated between 0 and 255 to avoid the wrapping effect of int8 values
- The modification to the decision variables appears as:
- $$P_{m.n}' = \begin{cases} 255, & P_{m.n} d_i > 255 \\ 0, & P_{m.n} d_i < 0 \\ P_{m.n} d_i, & otherwise \end{cases}$$



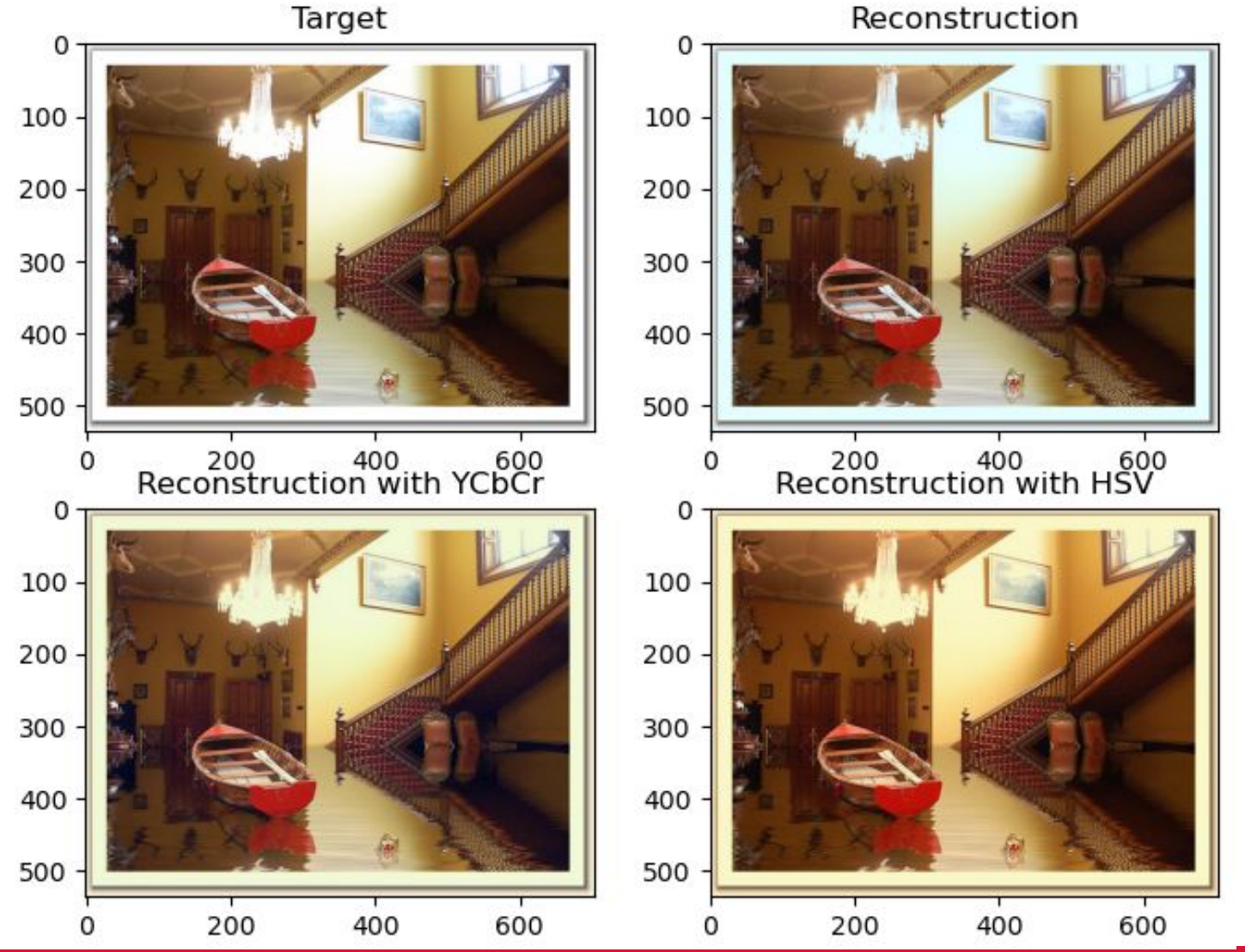
Workflow:

- Add decision variable modifiers to scale degraded image
- Convert reference image and degraded image into L*a*b* format
- Take the Delta E value of the two images
- Modify the weights to minimize Delta E
- Use the optimized weights to see the changed image

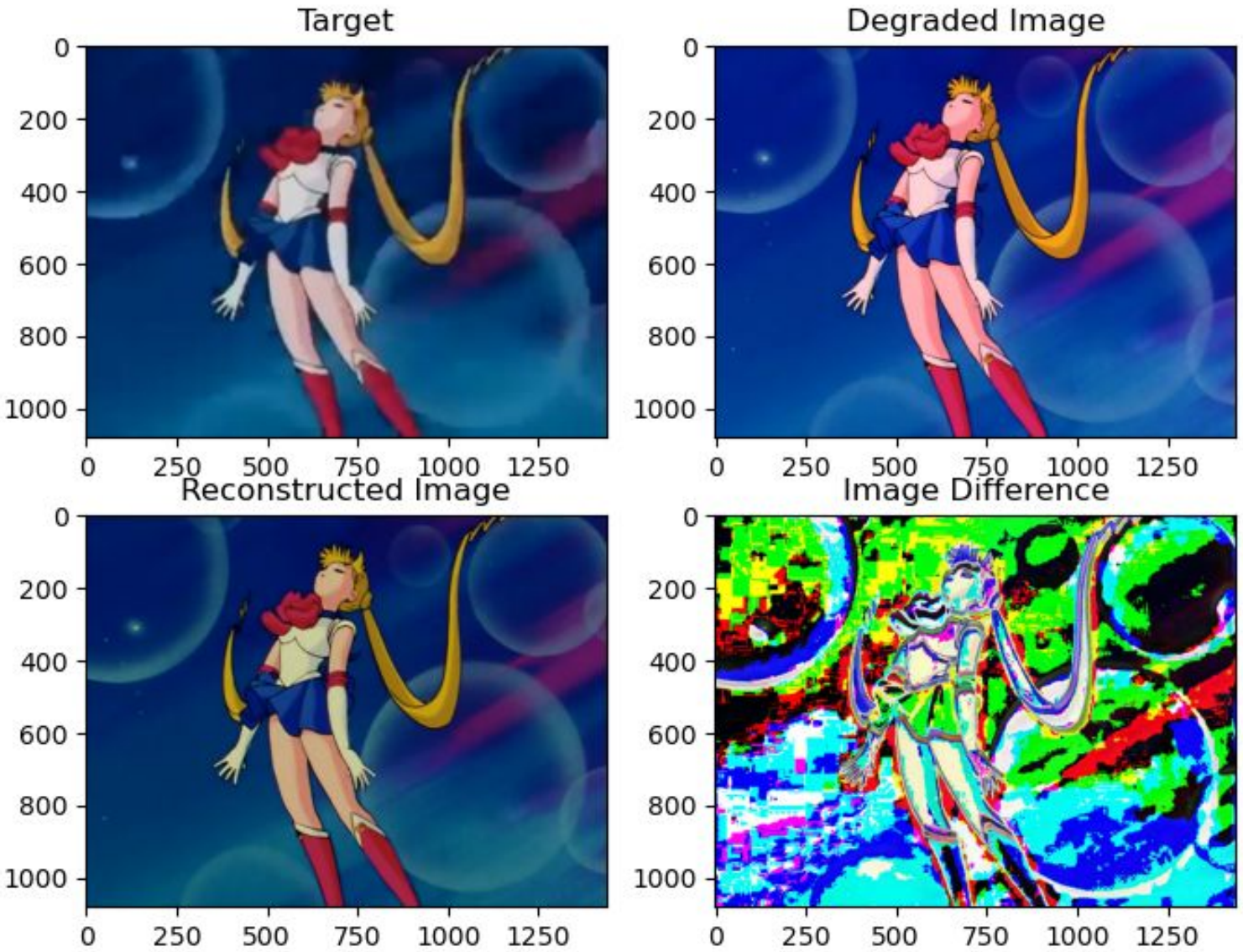
Results

Test Images

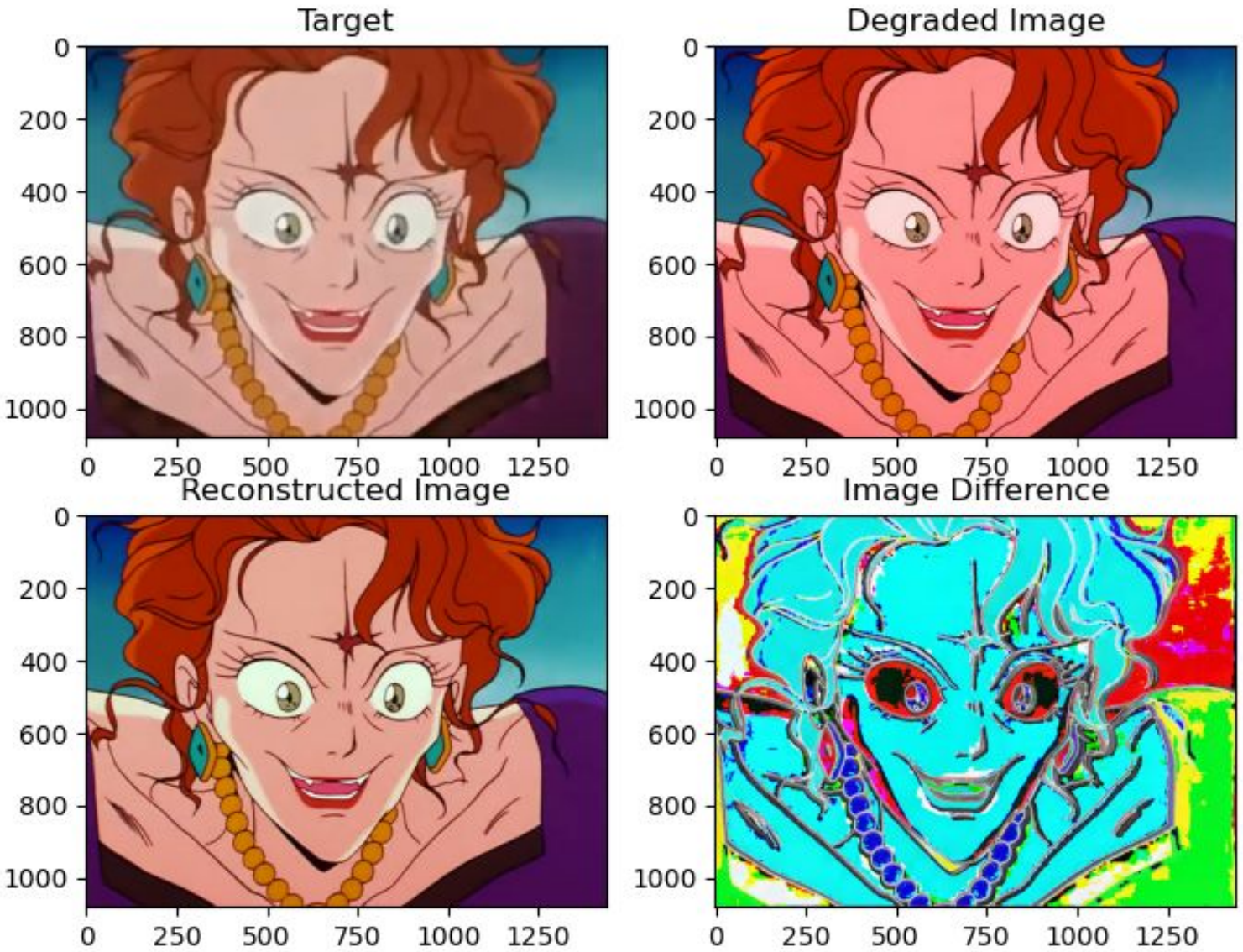
Test of Different Models



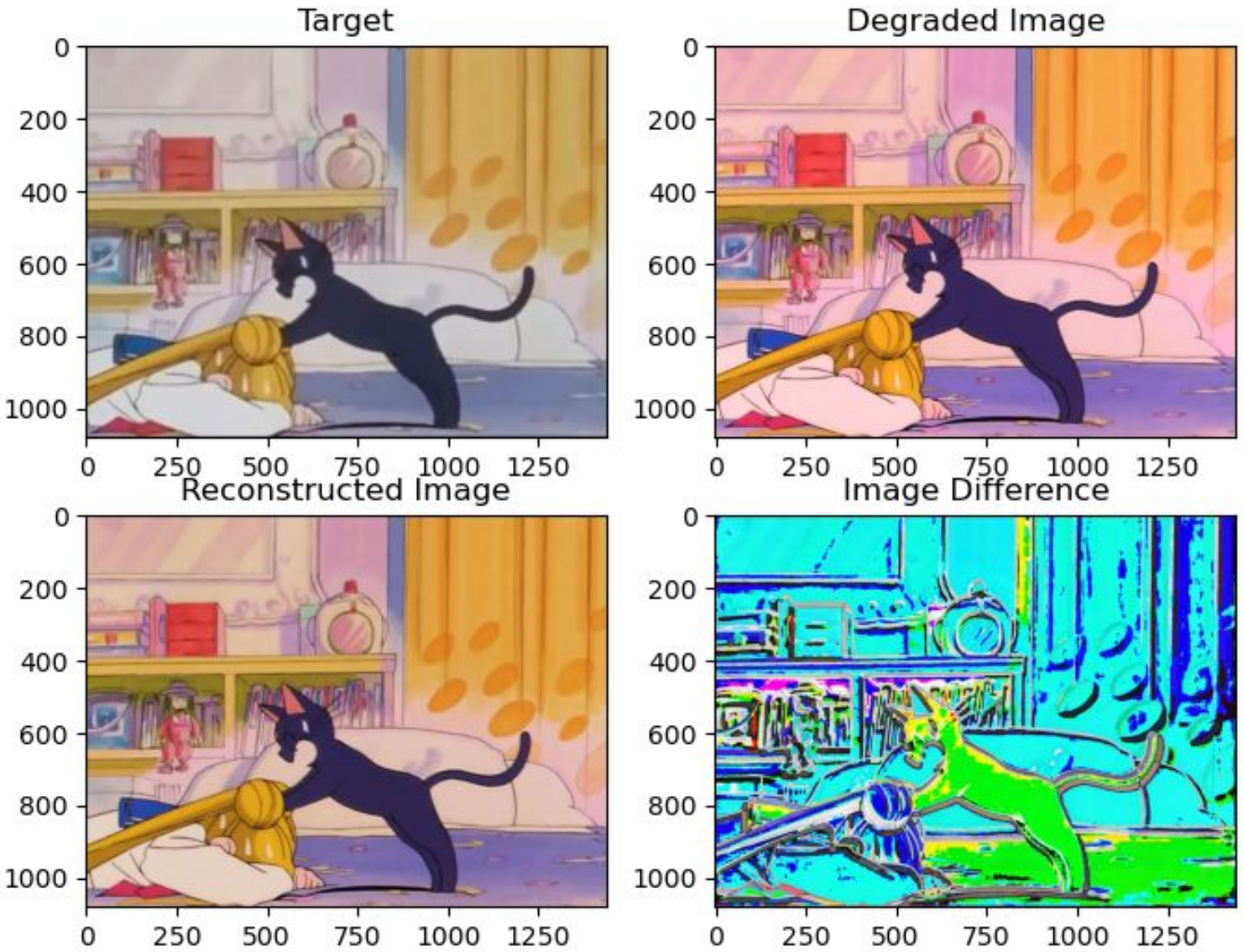
Results


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

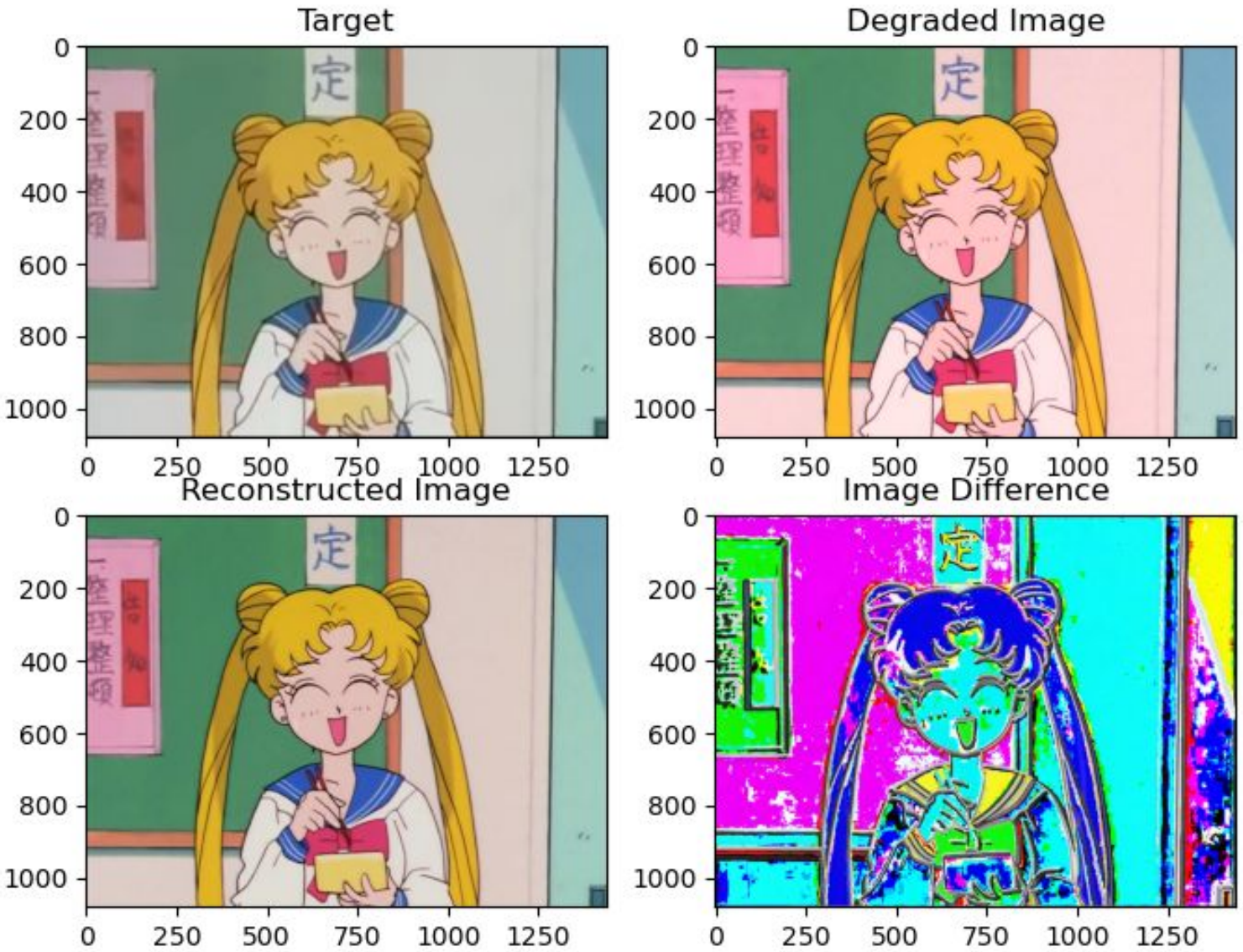
Results


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

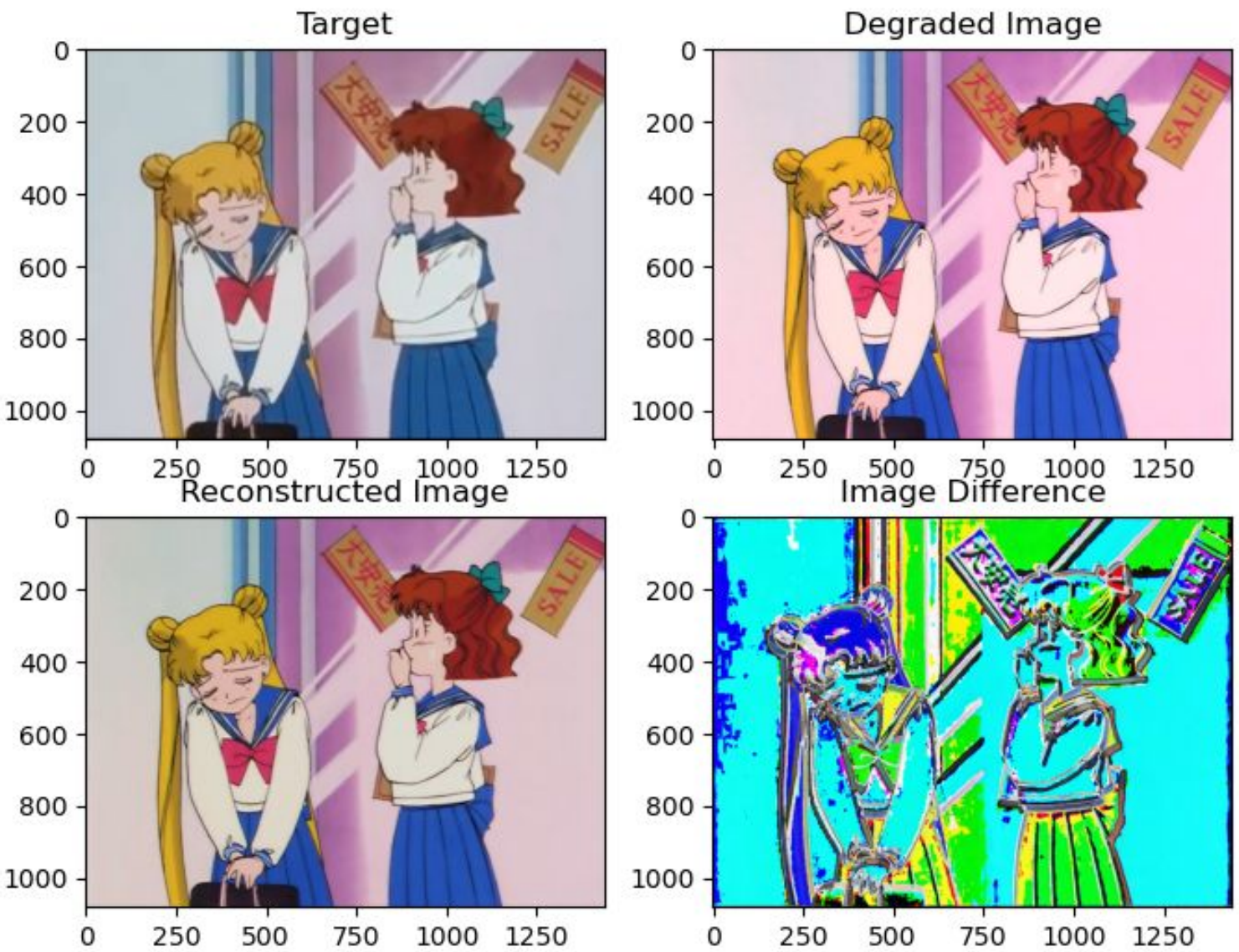
Results


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

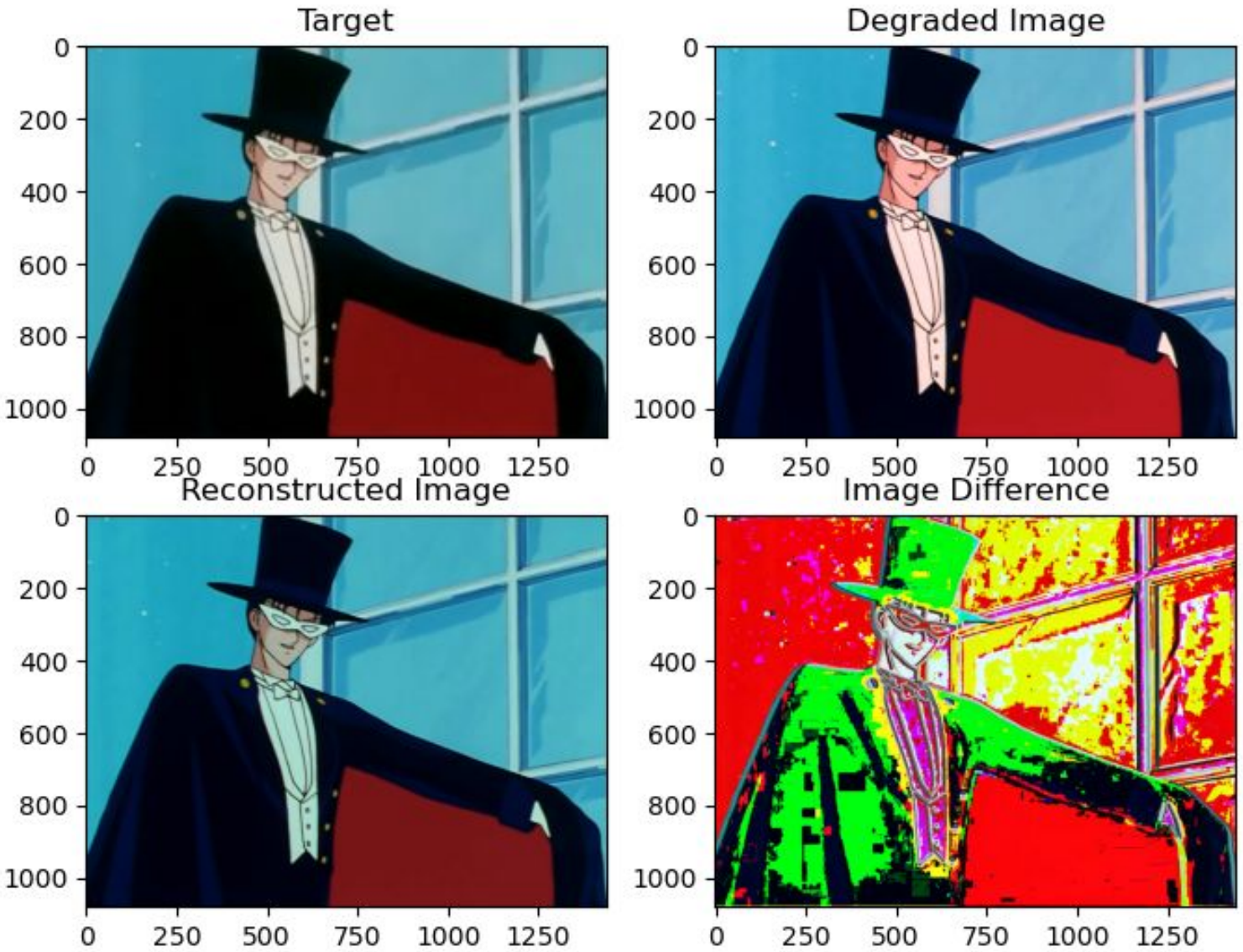
Results


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

Results


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250
SALE

Results


Target
Degraded Image
Reconstructed Image
Image Difference
0
200
400
600
800
1000
0
250
500
750
1000
1250

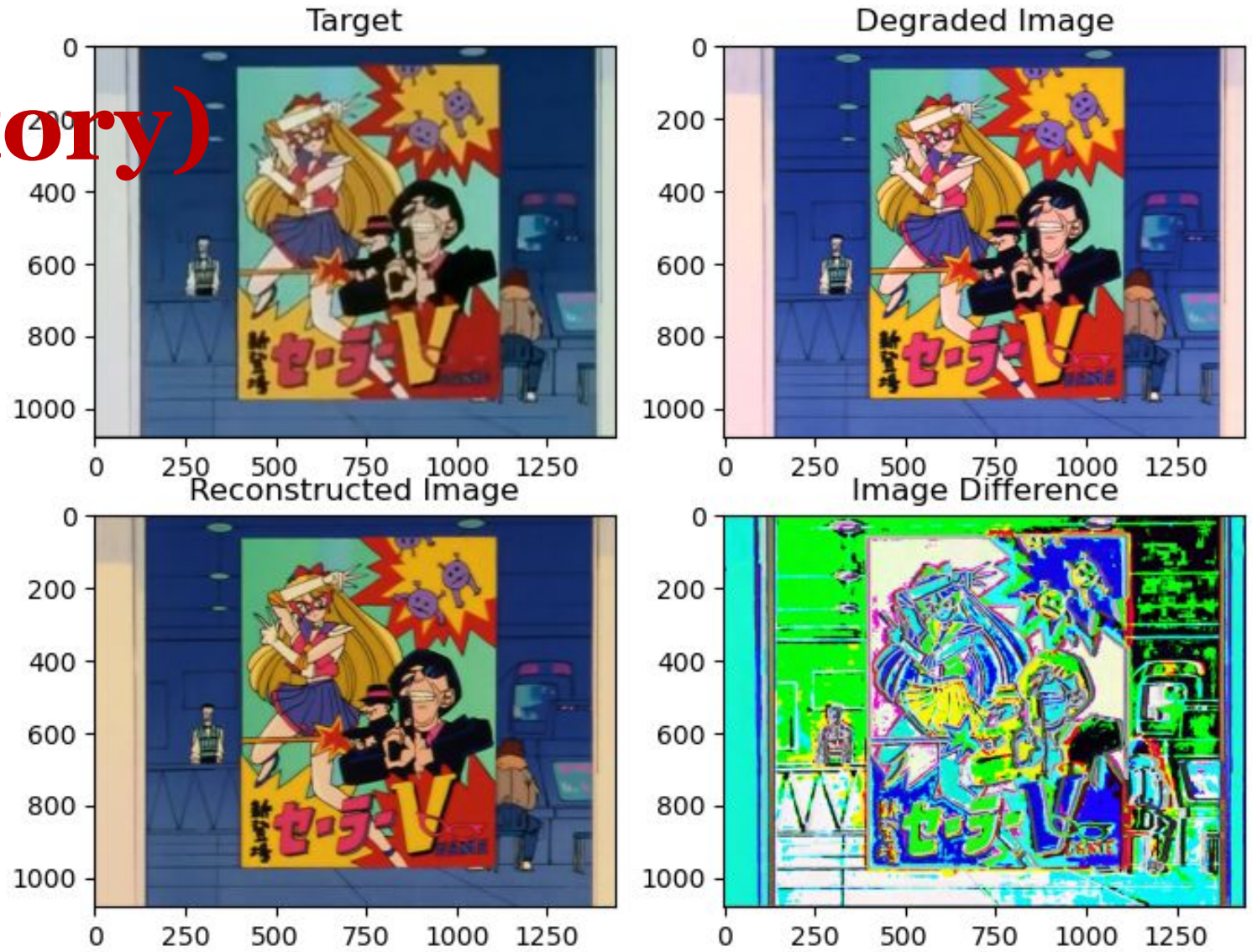
Frame	Weights	Original Delta E	New Delta E	% Improvement
1	[0.79217436 0.98040873 0.72790101]	24536468.38	14353988.05	41.50%
2	[0.91256059 1.16861554 1.05917191]	19089210.99	16502820.68	13.55%
3	[0.90921162 0.98743723 0.90100397]	15508966.79	13265572.21	14.47%
4	[0.88787144 0.96214723 0.91436317]	13785908.12	11078952.46	19.64%
5	[0.86945657 0.95714235 0.89454372]	16893338.66	12857371.25	23.89%
6*	[0.91209662 0.91754758 0.76051037]	22802316.48	18045120.06	20.86%
7*	[1.14883503 0.88248345 0.74120477]	14962440.53	10819630.11	27.69%
8	[0.65921003 0.97459486 0.93325614]	14579397.28	9134499.865	37.35%

Final Statements

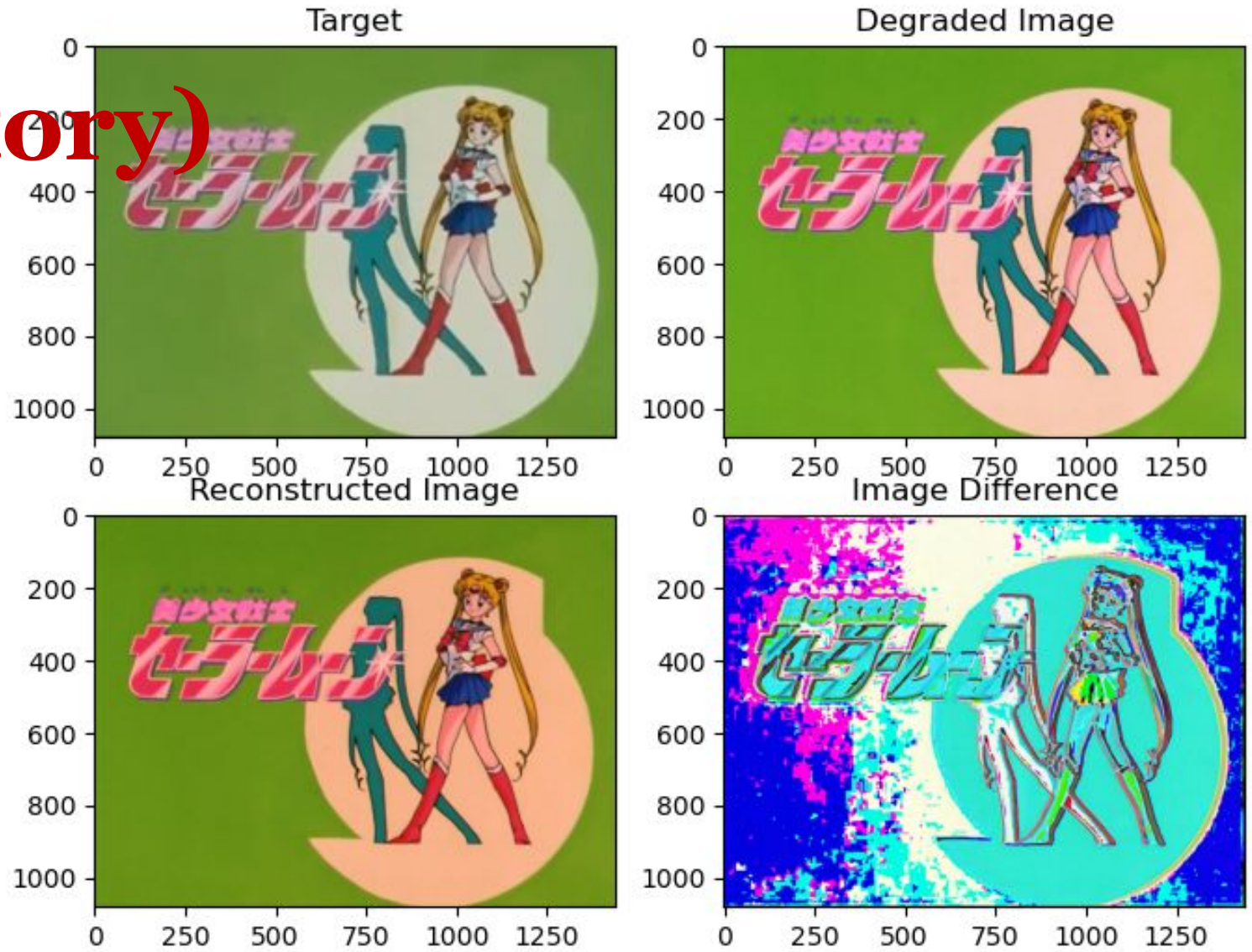
Results

- Am very satisfied with how optimization performed with restoring images, given that each decision variable was constant throughout the entire image, (much different from the standard)
- Even with images that did not have a high improvement percentage, qualitatively improved significantly
- There are a few exceptions . . .

Results, (Derogatory)

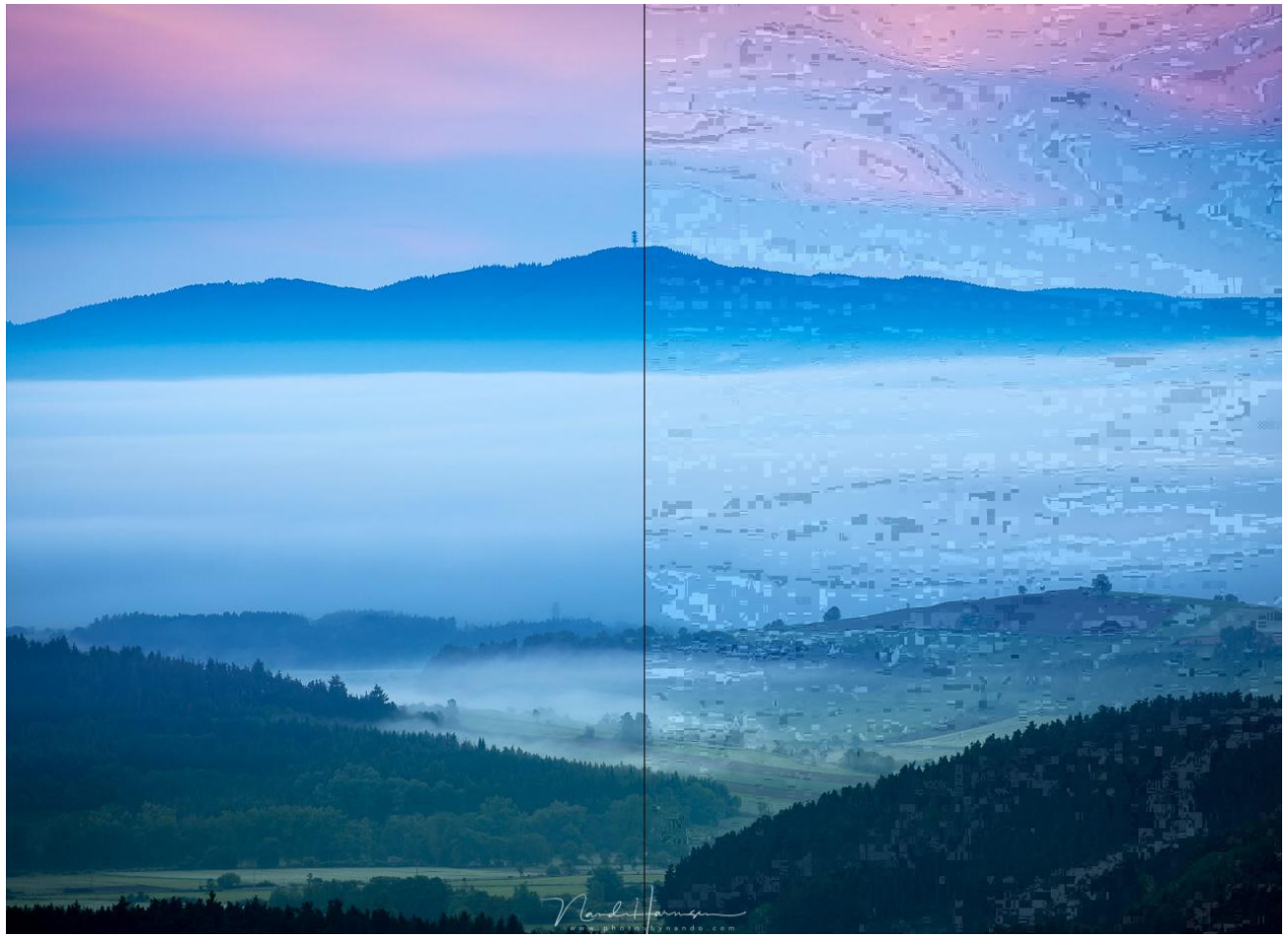


Results, (Derogatory)



JPEG Degradation

- Digital images are not exclusively immune to the process of image degradation, and it often happens because of the JPEG compression scheme
- As a JPEG continually gets saved, encoding and decoding process to reduce the size of the image will repeat, resulting in some data becoming irrecoverable



Best Course of Action?

- Still maintain physical media, but make sure to maintain good preservation practices
- Utilize more iterative models that take into psychovisual preferences into consideration, including the usage of edge detection and N-means clustering to extract objects of interest
- Store digital images in singular locations to avoid the process of degradation

References

[1] https://tedium.co/2024/03/02/why-physical-objects-decay/

[2] https://news.yahoo.com/sailor-moon-fans-are-surprised-to-learn-the-reason-behind-the-animes-aesthetic-pink-hue-it-wasnt-supposed-to-look-like-that-005916237.html

[3] https://unwritten-record.blogs.archives.gov/2016/01/11/film-preservation-101-why-are-old-films-sometimes-pink/

[4] http://www.brucelindbloom.com/index.html?Eqn_DeltaE_CIE2000.html

[5] https://hajim.rochester.edu/ece/sites/gsharma/ciede2000/ciede2000noteCRNA.pdf

[6] https://www.handprint.com/HP/WCL/color7.html#CIELUV

[7] https://colour.readthedocs.io/en/latest/generated/colour.delta_E.html

[8] https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html

References

Images:

[1] Stack of DVD's, https://commons.wikimedia.org/wiki/File:DVD_Stack.JPG

[2] "Duffless", Season 4, Episode 16, *The Simpsons*, Disney

[3] https://www.imdb.com/title/tt0103369/

[4] *Sailor Moon*, Toei Animation and Viz Media

[5] https://thedarkroom.com/slide-film-vs-color-negative-film/

[6] RGB Cube, https://commons.wikimedia.org/wiki/File:RGB_Cube_Show_lowgamma_cutout_b.png

[7] HSV Mapped to Cylinder, https://en.m.wikipedia.org/wiki/File:HSV_color_solid_cylinder.png

[8] Animated YCbCr Color Space, https://commons.wikimedia.org/wiki/File:YCbCr.GIF

[9] Flooded House, public domain

[10] JPEG quality degradation, https://fstoppers.com/education/about-jpeg-images-and-their-quality-degradation-435235

Thank You